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Effect of saline irrigation temperature difference on postoperative acute pain and hypothermia during biportal endoscopic spine surgery

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Severance

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Introduction

❑ Biportal endoscopic spine surgery (BESS)

- Minimally invasive surgery
- Using two independent portals
- Large volume of irrigation fluid

❑ Irrigation fluid in BESS

- Clarifying the operation field view
- Washout the bone or soft tissue debridement
- Reduce the possibility of operation site infection
- Affect on core body temperature

Introduction

❑ Knee arthroscopy

- When **irrigation fluids are warmed**, the risk of **hypothermia is less** and the risk of postoperative **shivering is reduced**.

Victoria et al. Arthroscopy. March 2018

❑ Shoulder arthroscopy

- Low level of inflammatory cytokine(IL-6)
- **More often hypothermia event** when using **room temperature** irrigation fluid

Xiaoyun et al. Arch Orthop Trauma Surg. 2015 Aug;135(8):1131-9

❑ Endoscopic urethral treatment

- **Isothermic irrigation decrease** both the reduction in core temperature and the incidence of shivering.

Mehmet et al. Urology Journal, Vol. 17 No. 1 (2020), 26 January 2020

Purpose of this study

- ❑ Effectiveness of using warm irrigation saline for spine endoscopic surgery ?
 - No consensus for spine endoscopic surgery



Material and methods

- ❑ Included cases from **2022-07-01 to 2023-07-31 (1years)**

- ❑ **Single hospital, single surgeon**
- ❑ **Total 60 cases are enrolled**
 - 5 cases are excluded due to exclusion criteria (other inflammatory event)
 - Only for primary BESS decompression or discectomy

- ❑ **Statistical analysis**
 - Student's t-test for continuous, normally distributed data
 - Chi-squared test for categorical data
 - The collected data were analyzed in a timely order and chronologically

Material and methods

❑ Irrigation saline temperature control

- 3000cc saline bag
- Using heating cabinet warm up the saline up to 38 degree
- **Room temperature(18~20 °C) saline(3000cc) + Warm(38 °C) saline(3000cc)**
 - Mixed up before get into the body
 - Minimize the thermal damage
- **No water pump**
 - ➔ **Using irrigation tower (2m height, natural drainage)**

Material and methods

❑ Room temperature

- Strictly controlled by air conditioning system (18~20 °C)
- Patients were warmed up to maintain the core temperature routinely by warming blanket

❑ Evaluated Laboratory finding, Clinical outcome

- Lab – Hb, WBC, ESR, CRP, IL-6, Serum amyloid A(SAA), CK-MB, CK
- Clinical outcomes – VAS, Body temperature(Hypothermia event)
 - Preoperative / Immediate postoperative
 - 12 hours after surgery
 - POD 1 day / POD 2 days / POD 3 days

Results

- **Total 55 cases (Warm 27 cases, Cold 28 cases)**
 - No significant difference in operative time, Estimated blood loss

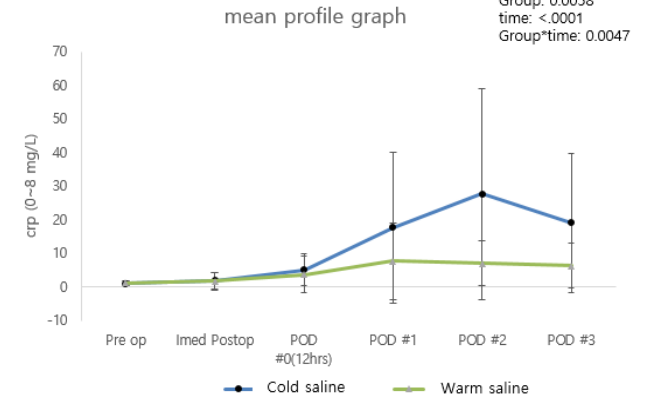
Table 1 Demographic data				
	Warm saline group	Cold saline group	Total	P-value
Sex (M: F)	13: 14	16: 12	29: 26	0.0781
Age (years, mean $\pm$ SD)	52.67 $\pm$ 10.61	55.36 $\pm$ 17.87	54.04 $\pm$ 14.69	0.4989
Height (cm ² , mean $\pm$ SD)	168.10 $\pm$ 7.90	165.56 $\pm$ 9.20	166.80 $\pm$ 8.60	0.2779
Weight (kg, mean $\pm$ SD)	77.70 $\pm$ 13.86	72.22 $\pm$ 16.45	74.91 $\pm$ 15.34	0.1880
BMI (kg/m ² , mean $\pm$ SD)	27.39 $\pm$ 3.71	26.08 $\pm$ 3.84	26.72 $\pm$ 3.80	0.2033
Operative time (min, mean $\pm$ SD)	112.89 $\pm$ 20.04	113.11 $\pm$ 20.84	113.00 $\pm$ 20.26	0.9686
EBL (cm ³ , mean $\pm$ SD)	82.96 $\pm$ 29.72	83.75 $\pm$ 26.30	83.36 $\pm$ 27.77	0.9175
BMI: body mass index				
EBL: estimated blood loss				
* P-value < 0.05 indicates a significant difference between both groups				

Results

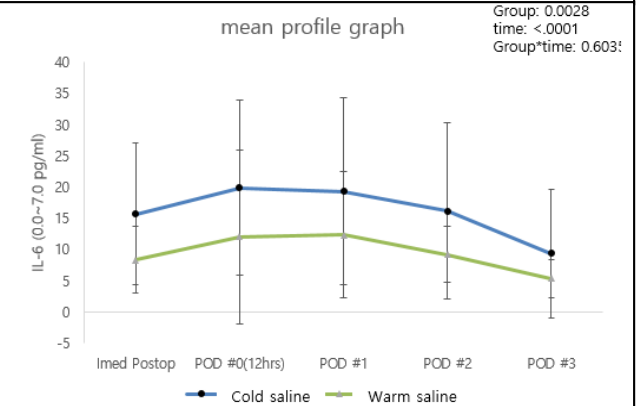
Table 2 Laboratory data of patients in the cold and warm saline groups

	Time	Cold saline group mean (SE)	Warm saline group mean (SE)
Hb (13.0–17.4 g/dL)	Preop	14.45 (0.279)	13.77 (0.284)
	Immediate Postop	13.83 (0.302)	13.68 (0.307)
	POD #0 (12 h)	13.35 (0.312)	13.18 (0.317)
	POD #1 (24 h)	12.97 (0.309)	12.75 (0.314)
	POD #2 (48 h)	12.65 (0.284)	12.18 (0.289)
	POD #3 (72 h)	12.88 (0.295)	12.30 (0.300)
	<i>p</i> =0.3617		
CK-MB (0.0–5.0 ng/ml)	Immediate Postop	2.39 (0.344)	2.54 (0.350)
	POD #0 (12 h)	3.75 (0.631)	3.07 (0.642)
	POD #1 (24 h)	3.31 (0.620)	3.51 (0.631)
	POD #2 (48 h)	2.33 (0.382)	2.80 (0.389)
	POD #3 (72 h)	1.98 (0.322)	2.14 (0.328)
	<i>p</i> =0.9216		
CK (55–289 IU/L)	Immediate Postop	135.89 (13.051)	128.59 (13.290)
	POD #0 (12 h)	229.00 (25.893)	164.44 (26.368)
	POD #1 (24 h)	226.61 (25.810)	189.59 (26.284)
	POD #2 (48 h)	201.25 (26.228)	189.93 (26.709)
	POD #3 (72 h)	165.93 (20.690)	156.63 (21.070)
	<i>p</i> =0.384		
WBC (4000–10,800 / μ L)	Preop	6341.63 (312.298)	6537.41 (318.029)
	Immediate Postop	8214.64 (580.805)	9195.19 (591.463)
	POD #0 (12 h)	8444.29 (394.415)	8961.85 (401.653)
	POD #1 (24 h)	7969.29 (424.741)	8644.44 (432.535)
	POD #2 (48 h)	6992.86 (388.749)	6425.93 (395.883)
	POD #3 (72 h)	6177.86 (292.323)	6330.74 (297.687)
	<i>p</i> =0.4622		
ESR (0–20 mm/hr)	Preop	7.04 (2.429)	5.00 (2.474)
	Immediate Postop	5.43 (1.364)	5.41 (1.389)
	POD #0 (12 h)	5.96 (2.006)	6.56 (2.043)
	POD #1 (24 h)	8.71 (3.038)	8.74 (3.093)
	POD #2 (48 h)	13.75 (2.565)	8.59 (2.612)
	POD #3 (72 h)	15.18 (3.164)	11.04 (3.222)
	<i>p</i> =0.5591		

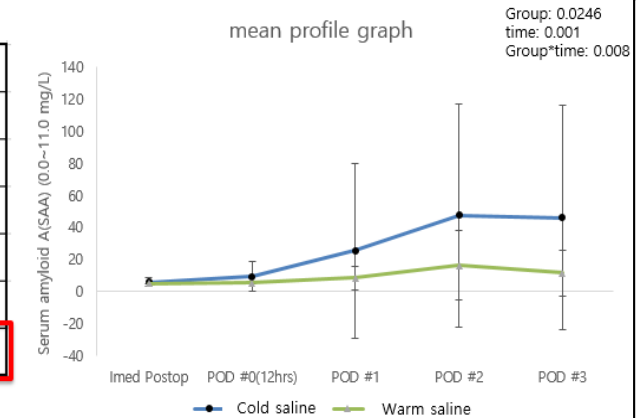
	Time	Cold saline mean(SE)	Warm saline mean(SE)
crp (0~8 mg/L)	Pre op	1.057(0.154)	1.081(0.157)
	Imed Postop	1.800(0.500)	1.659(0.509)
	POD #0(12hrs)	5.086(0.958)	3.607(0.976)
	POD #1(24hrs)	17.671(3.387)	7.581(3.449)
	POD #2(48hrs)	27.639(4.328)	6.967(4.407)
	POD #3(72hrs)	18.982(2.923)	6.252(2.977)
	<i>p</i> = 0.0058 < 0.05		



	Time	Cold saline mean(SE)	Warm saline mean(SE)
IL-6 (0.0~7.0 pg/ml)	Imed Postop	15.712(1.695)	8.320(1.726)
	POD #0(12hrs)	19.916(2.642)	12.012(2.691)
	POD #1(24hrs)	19.350(2.431)	12.339(2.476)
	POD #2(48hrs)	16.183(1.990)	9.230(2.026)
	POD #3(72hrs)	9.336(1.434)	5.334(1.460)
	<i>p</i> = 0.0028 < 0.05		



	Time	Cold saline mean(SE)	Warm saline mean(SE)
Serum amyloid A (SAA) (0.0~11.0 mg/L)	Imed Postop	6.000(0.334)	5.170(0.340)
	POD #0(12hrs)	9.529(1.298)	5.563(1.322)
	POD #1(24hrs)	25.436(7.440)	8.600(7.576)
	POD #2(48hrs)	47.500(9.826)	16.296(10.007)
	POD #3(72hrs)	46.189(9.648)	11.681(9.825)
	<i>p</i> = 0.0246 < 0.05		

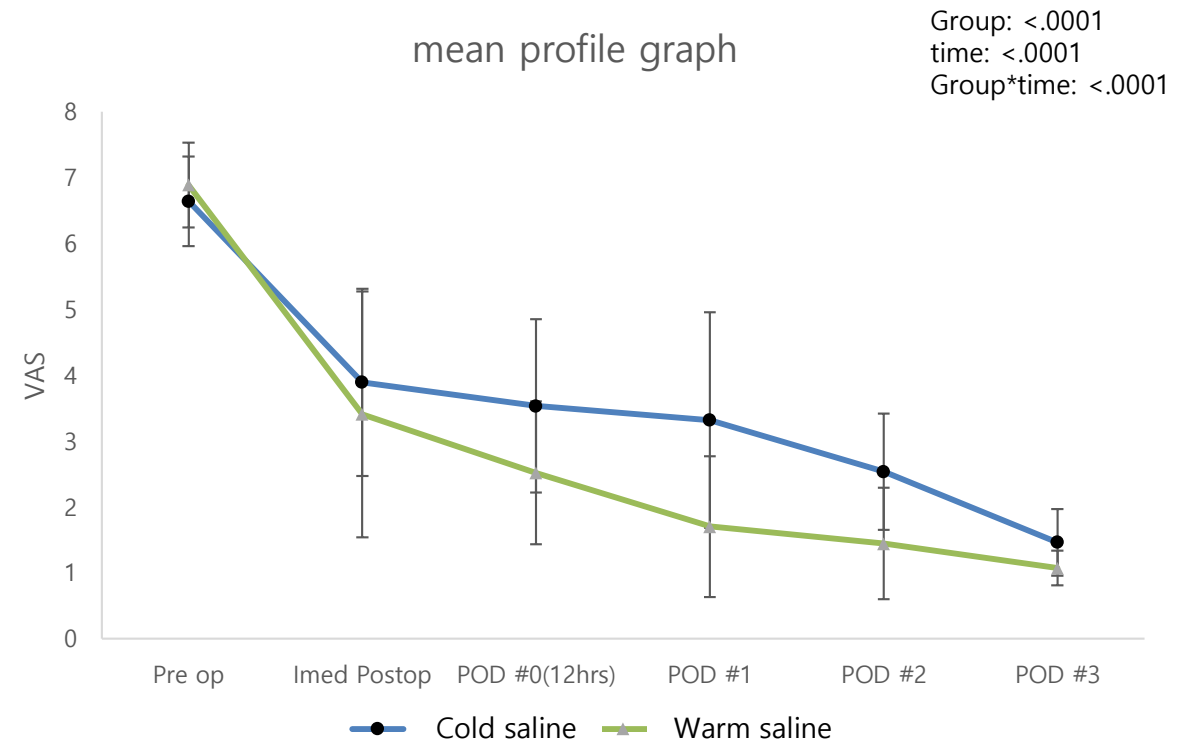


Result

□ Clinical outcome – VAS

- Warm saline group had lower VAS. ($p < 0.0001$)

	Time	Cold saline mean(SE)	Warm saline mean(SE)
VAS	Pre op	6.643(0.125)	6.889(0.127)
	Imed Postop	3.893(0.313)	3.407(0.319)
	POD #0(12hrs)	3.536(0.229)	2.519(0.233)
	POD #1(24hrs)	3.321(0.262)	1.704(0.267)
	POD #2(48hrs)	2.536(0.163)	1.444(0.166)
	POD #3(72hrs)	1.464(0.077)	1.074(0.078)
$p < 0.0001$			



Results



Body temperature

- Significant different between two groups
- Except for pre-operative phase



Hypothermia incidence ratio

- Did not occur in both groups until 30 minutes after operation
- Cold saline group
 - Incidence ratio ↑
 - Degree of recovery of temperature ↓

Table 4 Body temperature data and the incidence rate of hypothermia events in patients in the warm and cold saline groups

	Time	Cold saline group mean (SE)	Warm saline group mean (SE)	P-value
Body temperature (°C)	Preop	36.686 (0.047)	36.681 (0.047)	0.9494
	Op start	36.436 (0.032)	36.619 (0.033)	0.0002 *
	15 min	36.318 (0.040)	36.504 (0.041)	0.0021 *
	30 min	36.164 (0.040)	36.363 (0.041)	0.001 *
	45 min	36.029 (0.044)	36.285 (0.045)	0.0001 *
	60 min	35.943 (0.049)	36.163 (0.050)	0.0027 *
	75 min	35.841 (0.050)	36.042 (0.051)	0.0071 *
	90 min	35.759 (0.047)	35.980 (0.048)	0.0018 *
	105 min	35.674 (0.048)	35.882 (0.049)	0.0038 *
	120 min	35.620 (0.045)	35.835 (0.046)	0.0015 *
	Immediate postop	36.111 (0.047)	36.389 (0.048)	0.0001 *
	p-value = 0.0002 *			

Table 4 Body temperature data and the incidence rate of hypothermia events in patients in the warm and cold saline groups

	Time	Cold saline group mean (SE)	Warm saline group mean (SE)	P-value
Hypothermia incidence ratio (< 36°C)	Preop	0/28	0/27	
	Op start	0/28	0/27	
	15 min	0/28	0/27	
	30 min	5/28 (17.86%)	0/27	0.0136 *
	45 min	10/28 (35.71%)	2/27 (7.41%)	0.0063 *
	60 min	17/28 (60.71%)	5/27 (18.52%)	0.0004 *
	75 min	19/25 (76.00%)	10/24 (41.67%)	0.0066 *
	90 min	22/25 (88.00%)	14/24 (58.33%)	0.0037 *
	105 min	21/22 (95.45%)	15/21 (71.43%)	0.0195 *
	120 min	19/20 (95.00%)	12/18 (66.67%)	0.0131 *
	Immediate post OP	7/28 (25.00%)	3/27 (11.11%)	0.1723
	P-value = 0.0005 *			
Immediate postoperative shivering		2/28 (7.14%)	0/27 (0.00%)	

* P-value < 0.05 indicates a significant difference between the two groups



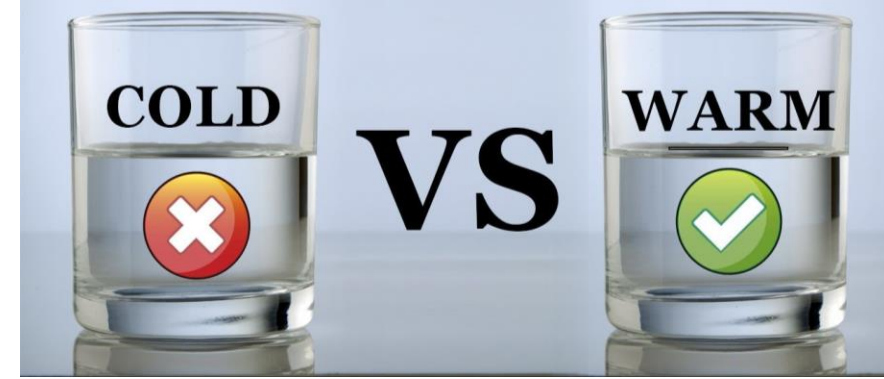
Limitation of study

- ❑ **Could not sufficiently control variables**
 - Number of levels / Diseases (stenosis, disc, etc)
 - Patients' underlying conditions

- ❑ **Different methods of measurement patients' core body temperature**
 - Esophageal stethoscope – During procedure
 - Digital tympanic thermometer – Post anesthesia care unit

- ❑ **Single surgeon performance**

Summary of results



❑ Laboratory findings (IL-6, SAA, CRP)

- The acute inflammatory cytokines such as **IL-6, SAA(serum amyloid A) and CRP had lower level** at “Warm saline group”

❑ Clinical outcomes (VAS)

- The **Back VAS was more lower** at “Warm saline group” compared with “Cold saline group”.

❑ Complications (Hypothermia, shivering)

- The **hypothermia events were occurred more frequently** at “Cold saline group”, compared with “Warm saline group”.

Conclusion

- ❑ Using warm saline irrigation could **reduce the acute inflammatory reaction.**
- ❑ The reduced acute inflammatory reaction could **affect the clinical outcome of BESS** which come out to be **lower VAS.**
- ❑ Using warm saline could **diminish the complications such as hypothermia or shivering.**
- ❑ **Using warm irrigation saline** have **lots of benefits** in BESS

This study is published in *European Spine journal*, 27 May 2024

Cited : <https://doi.org/10.1007/s00586-024-08322-6>

Thank you for listening



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